

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: plasmids and cosmids, 40 kb and smaller.
Species Used	Mouse, 3T3, embryo; Human, fibroblast; primary cell lines; Monkey, Vero, kidney cells.		

Before the Pulse

Cell Growth Medium	DMEM + 10% Nu serum (GIBCO/BRL, Sigma, Flow Labs)	Growth Phase at Harvest	Passaged 24 hours prior to pulse.
		Pre-pulse Incubation	minimum

Wash Solution Not given

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	25 °C		
Electroporation Medium*	Opti medium (BRL)	Cuvette Gap	0.4 cm
Cell Density	10 (7) cells / pulse	Voltage	0.220 kV
Volume of Cells	0.4 ml	Field Strength	0.55 kV/cm
DNA Concentration	10 to 20 µg / pulse	Capacitor	960 µF
DNA Resuspension Buffer	20 µl TE (10 mM Tris, 1 mM EDTA, pH 8.0) / pulse	Resistor	(Pulse Controller) Ω none
Volume of DNA	20 µl	Time Constant	30 msec
After the Pulse			
Outgrowth Medium	DMEM + 10% Nu serum		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	24 hours
Selection Method or Assay Used	b-gal expression: X-gal staining. Replication of Herpes origin of replication between S plasmids
Electroporation Efficiency	30 to 50% cells transfected
Per Cent Survival	60%

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Survey Number

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